1 QUESTION

RESEARCH QUESTIONS

Q1 · Proportion

Is the proportion of students who reported feeling sad or hopeless different from **0.30**?

Q2 · Mean

Is the mean body weight of students different from **68.0 kg**?

DATA & VARIABLES

CSV YRBS_2007 · 14,841 rows

CSV yrbs_combined_processed · 14,042 rows

SadOrHopeless
n = 13,845

Weight (kg)
n = 13,862

ADDITIONAL EDA SET — SHARED N = 13,120

CurrentCigaretteUse
n = 13,120

HowOldAreYou
n = 13,120

With **SadOrHopeless** included, the shared valid n is **13,120**.

01

Population Proportion · **SadOrHopeless**

EDA HIGHLIGHT

SAMPLE PROPORTION

30.0%

4,153 / 13,845

reported SadOrHopeless

BINARY RECODED DISTRIBUTION

Yes 30.0%

No 70.0%

METHOD & RESULT

ONE-SAMPLE Z-TEST

H_0: p = 0.30
H_1: p ≠ 0.30

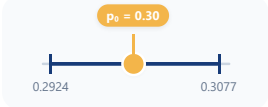
α = 0.05 · Two-sided · Wald 95% CI

$z = (\hat{p} - p_0) / \sqrt{p_0(1-p_0)/n}$

Decision
Fail to reject H_0 — estimate essentially equal to 0.30.

RESULT

$\hat{p}$	**0.2999 (≈ 30.0%)**
95% CI	**[0.2924, 0.3077]**
z	**−0.0093**
p-value	**0.9926**



02

Population Mean · **Weight (kg)**

EDA HIGHLIGHT

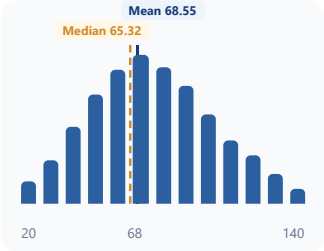
SAMPLE MEAN

68.55

Median 65.32 · SD 16.99

Right-skewed distribution · 447 outliers

WEIGHT DISTRIBUTION · KG



METHOD & RESULT

ONE-SAMPLE T-TEST

H_0: μ = 68.0
H_1: μ ≠ 68.0

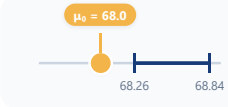
α = 0.05 · Two-sided · 95% CI for μ

$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$

Decision
Reject H_0 — mean weight is statistically above 68.0 kg.

RESULT

$\bar{x}$	**68.55 kg**
95% CI	**[68.26, 68.84]**
t	**3.7007**
p-value	**0.0002**



Conclusion

OVERALL SUMMARY & INSIGHTS

Prevalence of feeling sad/hopeless is **indistinguishable from the 30% benchmark** ($\hat{p}$ = 0.300, p = .99) — fail to reject H_0.

Mean body weight is **significantly higher than the 68.0 kg benchmark** ($\bar{x}$ = 68.55, p < .001) — reject H_0, though the **0.55 kg** gap is statistically but not practically large.

Skew and 447 outliers warrant cautious weight interpretation.

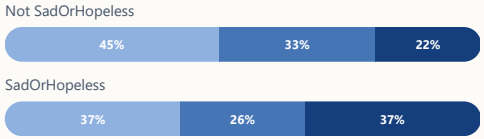
EXPL. · Q1 SMOKERS VS. NON-SMOKERS BY AGE

Is SadOrHopeless more common among students who reported smoking on at least 1 day in the past 30 days, and does this association vary across age groups?

Smoker − non-smoker SadOrHopeless gap is positive across every age group.

EXPL. · Q2 SMOKING FREQUENCY BY SADORHOPELESS

Among current smokers, does smoking frequency appear to differ by SadOrHopeless status?



Among current smokers, SadOrHopeless teens skew toward more frequent smoking.